Customer Segmentation & Purchase Behavior Analysis

Total Customers

350

Sum of Custome

Total Revenue

295.88K

Sum of Total Sp

Average Rating

4.02

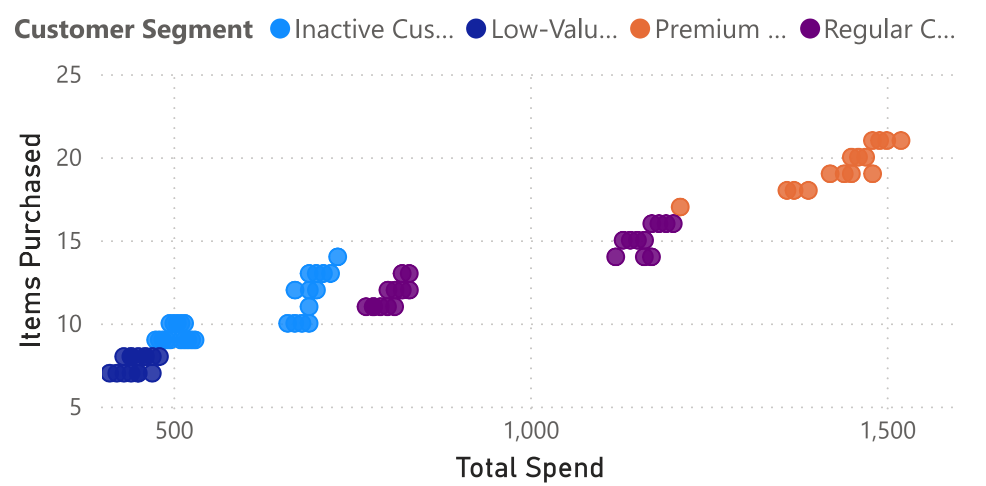
Average Rating

Average Spend

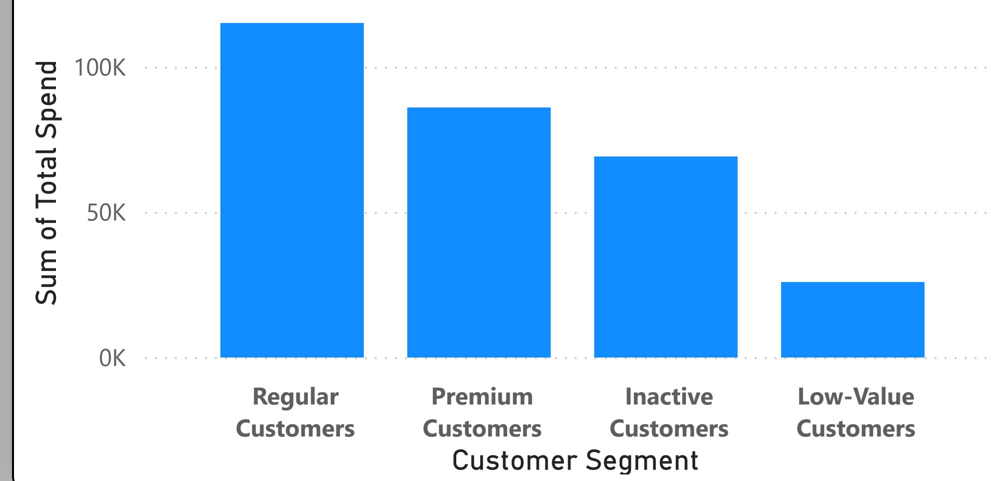
845.38

Average Spend

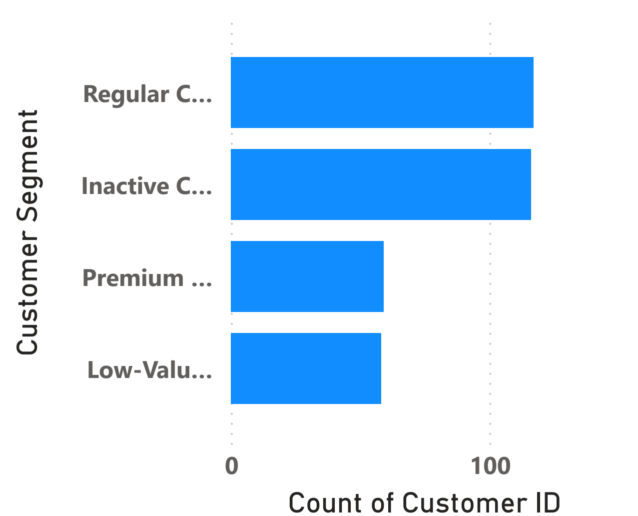
Customer Segments



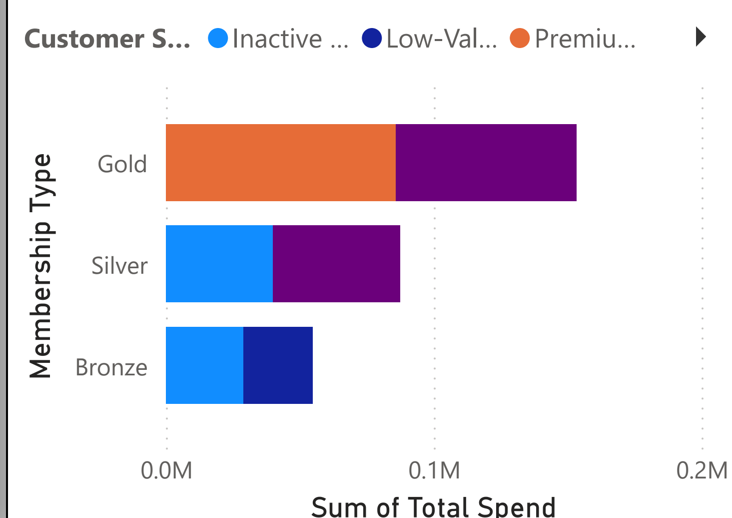
Sum of Total Spend by Customer Segment



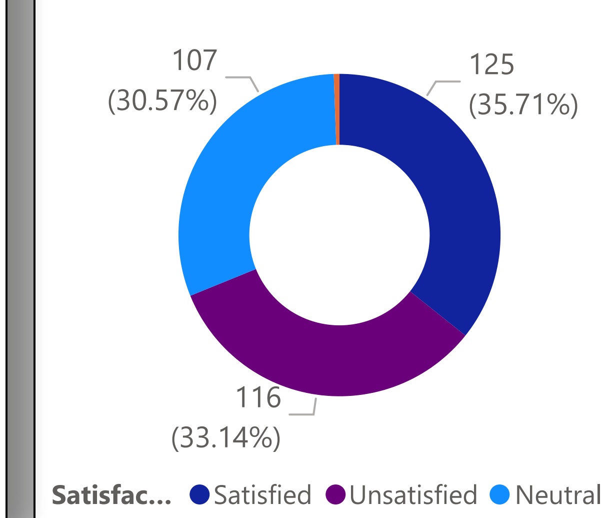
Count of Customer ID by Customer Segment



Sum of Total Spend by Membership Type and Customer Segment



Count of Customer ID by Satisfaction Level



Customer_Segmentation_Analy ×

colab.research.google.com/drive/1T1OqYdCja25aurUHxO4vg-q1zYHGksQO#scrollTo=Q-UPhiOwWg9x

Customer_Segmentation_Analysis.ipynb

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```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
```

```python
df = pd.read_csv("E-commerce Customer Behavior.csv")
```

```python
df.head()
```

	Customer ID	Gender	Age	City	Membership Type	Total Spend	Items Purchased	Average Rating	Discount Applied	Days Since Last Purchase	Satisfaction Level
0	101	Female	29	New York	Gold	1120.20	14	4.6	True	25	Satisfied
1	102	Male	34	Los Angeles	Silver	780.50	11	4.1	False	18	Neutral
2	103	Female	43	Chicago	Bronze	510.75	9	3.4	True	42	Unsatisfied
3	104	Male	30	San Francisco	Gold	1480.30	19	4.7	False	12	Satisfied
4	105	Male	27	Miami	Silver	720.40	13	4.0	True	55	Unsatisfied

Next steps: Generate code with df | New interactive sheet

```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 350 entries, 0 to 349
Data columns (total 11 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Customer ID               350 non-null    int64
 1   Gender                    350 non-null    object
 2   Age                       350 non-null    int64
 3   City                      350 non-null    object
 4   Membership Type           350 non-null    object
 5   Total Spend               350 non-null    float64
 6   Items Purchased           350 non-null    int64
 7   Average Rating            350 non-null    float64
 8   Discount Applied          350 non-null    bool
 9   Days Since Last Purchase  350 non-null    int64
 10  Satisfaction Level        348 non-null    object
dtypes: bool(1), float64(2), int64(4), object(4)
memory usage: 27.8+ KB
```

```python
df.columns
```

```
Index(['Customer ID', 'Gender', 'Age', 'City', 'Membership Type',
       'Total Spend', 'Items Purchased', 'Average Rating', 'Discount Applied',
       'Days Since Last Purchase', 'Satisfaction Level'],
```

Variables Terminal

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